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PROJECT TITLE

PREDICTIVE LEARNING SYSTEM FOR MATHEMATICS FORM 4 AND
FORM 5

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PREDICTIVE LEARNING SYSTEM FOR MATHEMATICS FORM 4 AND
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PROJECT CATEGORY

SYSTEM DEVELOPMENT PROJECT

by

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ABSTRACT

Malaysia's education sector is rapidly adopting a digital transformation to align with the agenda of the Fourth Industrial Revolution (4IR), especially when it comes to teaching Mathematics. Improving the current education system requires moving away from the "one-size-fits-all" approach as its severe limitations are highlighted in the high failure rates of Sijil Pelajaran Malaysia (SPM) examination. Therefore, there is an urgent need for a digital educational system that can tailor content to any student's level of comprehension. This paper proposes a Predictive Learning System focusing on Form 4 and 5 Mathematic subjects by integrating machine learning techniques. The system uses a hybrid model that combines a Random Forest (RF) and a Deep Neural Network (DNN) to classify students into Low, Medium, and High mastery levels based on their academic and behavioral data. The technology provides highly customized learning experiences by dynamically modifying question complexity while accounting for response times. Furthermore, it will also include an interactive chatbot that helps students solve problems without disclosing the answers, encouraging "on-the-spot" critical thinking. Ultimately, this paper presents an educational tool that aligns with the current foundational technologies in Malaysia and offers policymakers practical ideas for integrating adaptive learning systems into schools.

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LIST OF ABBREVIATIONS

CSS	Cascading Style Sheets
4IR	Fourth Industrial Revolution
AI	Artificial Intelligence
DNN	Deep Neural Network
DSKP	Curriculum and Assessment Standard Document
EDA	Exploratory Data Analysis
ERD	Entity-Relationship Diagram
IoT	Internet of Things
ITS	Intelligent Tutoring System
KSSM	Kurikulum Standard Sekolah Menengah
LLM	Large Language Model
LMS	Learning Management Systems
ML	Machine Learning
NLP	Natural Language Processing
PDPA	Personal Data Protection Act
RF	Random Forest
SMOTE	Synthetic Minority Over-sampling Technique
SPM	Sijil Pelajaran Malaysia
STEM	Science, Technology, Engineering, and Mathematics

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

The Fourth Industrial Revolution represents a significant change in the global economic and social environment with the convergence of physical, digital and biological technologies. Foundational technologies include Artificial Intelligence (AI), Internet of Things (IoT), Blockchain, Cloud Computing, Big Data Analytics, Advanced Materials and Technologies are altering how current industries function. Thus, the modern workforce is in high demand for individuals who exhibit higher proficiency in complex problem-solving and data literacy. Consequently, a strong foundation in analytical skills that are deeply based in mathematics comprehension has evolved from being merely advantageous to crucial. For Malaysia to remain globally competitive, a transformation of the foundational frameworks is imperative in the hopes of cultivating these attributes into younger generations.

Malaysia's education sector is rapidly adopting a digital transformation by shifting towards the use of an intelligent system to align with the agenda of the Fourth Industrial Revolution (4IR). In a Malaysian student's academic journey, Sijil Pelajaran Malaysia (SPM) examination is regarded to be the most important examination as it serves as a gateway that determines the caliber of higher education pathways including Science, Technology, Engineering, and Mathematics (STEM) streams at the tertiary level. Therefore, with Mathematics being one of the core subjects in SPM, the need to score is imperative as it directly dictates comprehension and mastery in critical thinking while highlighting eligibility for STEM-related professions. Furthermore, failure in this subject jeopardizes academic portfolio, restricting them from pursuing highly sought after bachelor programs extending well beyond STEM fields.

However, even though the critical importance of Mathematics has been emphasized in the current syllabus, the failure rate is still considerably high. According to the SPM 2024 Examination Results Analysis Report officially released by the Ministry of Education Malaysia, 84,025 candidates, or 22.4% of the total cohort, failed Mathematics which is better in comparison to SPM 2023 by 0.8% where 85,700 failed the subjects. While there have been

improvements according to the recorded statistics, the failure is still considerably high. This infers that the crisis in mathematical proficiency is a persistent systemic problem rather than a singular anomaly.

Presently, students rely on conventional study materials such as physical workbooks, in-class instructions or general quiz platforms such as “Quizziz” and “Kahoot!”. While these platforms are useful for rote memorization and knowledge recall, they primarily offer standardised, surface-level question and answer form. The current issue that arises from these is that while resources are abundant, they are often fragmented. Students usually face the dilemma that when they are stuck on a hard question, the resources available does not explicitly explain their confusion due to the content being too general or it uses different terminologies that the students are unfamiliar with. Hence, this paper proposes a Predictive Learning System that dynamically alter its contents' difficulty parallel with the student's proficiency.

With the integration of a Predictive Learning System that advocates for the students behavioral and academic data, the system aims to offer a proactive approach in education while addressing the issues stated prior. Next, the system fosters a responsive learning environment that promotes in-depth comprehension rather than memorization. Ultimately, students who utilize the proposed system are expected to demonstrate an improvement in their academics, better critical thinking skills and improved on-the-spot performance.

1.2 Problem Description

Within the current educational framework there exists multiple problems that can be identified. Firstly, conventional study methods do not accommodate for the students' mastery level. According to Galaige & Torrisi-Steele (2017), the current educational framework follows a “one-size-fits-all” approach where the same content instructions are delivered to learners in the same format irrespective of their differences which leads to high failure rate. In tandem with this, Traditional LMS platforms often function as passive repositories of content, delivering identical modules to students regardless of their disparate mastery levels, learning paces, or cognitive styles. Valdemoro et. al. (2026) stated that, this lack of adaptability creates a significant "pedagogical mismatch," where struggling learners are overwhelmed by advanced concepts while high-performing students are disengaged by

repetitive foundational material, ultimately compromising learning efficiency and motivation. Secondly, there exists the lack of a dedicated digital learning platform that solely focuses on the current Mathematics KSSM (Kurikulum Standard Sekolah Menengah) Form 4 and Form 5 syllabus. Available educational systems are either too general or are incompatible with the current syllabus. Traditional intelligent systems lack real-time dynamic capability and reliance on probabilistic models fails to cater to student's needs (Villegas et. al., 2025). This inflexibility limits scalability and reduces long-term sustainability, especially in diverse educational contexts (Mannekote et. al., 2024). Finally, there is an absence of a real time guidance tool that can directly and concisely help students when they are stuck on a hard question. A significant number of students disengage from learning due to a lack of interactive, just-in-time, and adaptive support (Davar et. al., 2025). Next, conventional intelligent tutoring systems only show the final answer to them without showing the proper step-by-step calculation to get there especially in Mathematics. This is a major issue as it stunts the student's capability to independently figure out the complexity of the question on their own volition which ultimately impacts their critical thinking skills. Hence, this situation makes independent learning a not-engaging experience for the student and leads to students giving up earlier without having the determination to put in any effort.

Consequently, this infers that the abundance of available educational content is not directly proportionate to a student's success. For students to adapt and evolve beyond their limits, the current educational framework must also evolve beyond the static repositories. This creates an urgency of a predictive learning system that can dynamically challenge students in parallel to their understanding of the curriculum. Systems can analyze students' learning routine and cater content to them by utilizing state of the art technologies (Lin et. al., 2024) This adaptive approach improves students' overall engagement and comprehension by providing them with tailored learning experiences that cater to their strengths and weaknesses. Students who use AI-powered tutoring systems outperformed those who only rely on traditional classroom teaching in mathematics (Leddo & Garg, 2021). In addition, Son (2024) states AI tutoring systems successfully adjust to students' cognitive needs by employing scaffolding techniques, interactive feedback, and step-by-step problem-solving support. Hence, the core challenges within the current educational framework are the "Adaptability Gap", the "Curriculum Alignment Gap", and the "Guidance Gap".

- **The Adaptability Gap** : Traditional LMS platforms use a "one-size-fits-all"

approach, causing a "pedagogical mismatch" where students are either overwhelmed or disengaged.

- **The Curriculum Alignment Gap :** Available educational systems are often too general and incompatible with specific syllabi like Mathematics KSSM Form 4 and 5, limiting their sustainability.
- **The Guidance Gap :** Conventional tutoring systems lack interactive, real-time support and only provide final answers without step-by-step calculations, which stunts critical thinking.

1.3 Project Objectives

To ensure the systematic development of the proposed system and to meet the needs of the target users, this project establishes several key objectives covering data management, intelligent model development, and web platform implementation. Specifically, the objectives of this project are as follows:

- To develop a predictive learning system that classifies students' mastery levels into Low, Moderate, and High categories based on analysis of their academic performance profiles.
- To develop an adaptive quiz system that dynamically adjusts question difficulty according to students' response time and their classified mastery level.
- To develop an intelligent guidance chatbot module that provides step-by-step support and hints without revealing the final answers.
- To create a performance dashboard that visualizes students' mastery levels through graphical representations such as charts and performance analytics.

1.4 Scope of the Project

i. **Scope:** This project focuses on the development of an adaptive learning system for Form 4 and Form 5 Mathematics using Machine Learning (ML) techniques. The scope of the project encompasses the following components:

- **Academic Content:** The system is limited to Core Mathematics topics based on the national KSSM syllabus for Form 4 and Form 5. It utilizes a manually curated question bank where each question is labeled with a specific difficulty level to enable dynamic adaptation.
- **Functions:** A hybrid combining Random Forest (RF) and Deep Neural Network

(DNN)model will be developed to classify students' mastery levels into three categories: Low, Medium, and High.

- **Data Source:** Due to strict privacy constraints under the Personal Data Protection Act (PDPA) preventing access to real students' data, the project relies on a secondary dataset from Kaggle for model training and proof of concept.
- **Interactive Guidance:** The system integrates a Large Language Model (LLM) powered chatbot that acts as a virtual tutor. It provides step-by-step scaffolding and hints without directly revealing the final answers.

ii. **Target Audience:** The target audience of the proposed system is mainly Form 4 and Form 5 students who are preparing to undertake the Mathematics subject in the Sijil Pelajaran Malaysia (SPM) examination.

iii. **Specific Platform:** The proposed system is a web-based application developed using the Streamlit framework and the Python programming language. Its usage is specifically optimized for web browsers on desktop devices to ensure interface stability when students interact with the quiz modules and chatbot.

1.5 Constraints

- **Data Acquisition:** Strict compliance with the Personal Data Protection Act (2010) and institutional procedures prevented the use of primary student data for model training, requiring the use of secondary Kaggle datasets, which needed extensive pre-processing for contextual alignment.
- **Technology:** The system requires a stable Wi-Fi connection to facilitate quizzes, data processing, seamless chatbot interactions, and synchronization between the user interface and the backend system.
- **Resource Availability:** The question bank only consists of questions from SPM trial questions from the year 2023-2025

1.6 Project Stages

[illegible]

4	Design & Development (Chapter 3)	4 weeks																
	Report Writing Chapter 3																	
	Sketch (Draft design)																	
	User Interface Design																	
	Milestone : Draft 3 Submission	20/5/2026																
5	Prototype Implementation & Testing	5 weeks																
	Prototype Testing																	
	User Acceptance Testing (UAT)																	
	Milestone: Demonstrate Prototype Design & Functionality	TBD																
6	Documentation & Presentation	2 weeks																
	Compile FYP Technical Report (Ch 1-3, poster, and Turnitin report)																	
	Final FYP 1 report submission	1/6/2026																
	Milestone: Final Presentation at INNOVATEX	TBD																

1.7 Significance of the Project

This project offers significant benefits across two main areas:

- **For Students:** Enables early detection of learning gaps and provides personalized pathways for effective revision. The adaptive structure improves time management and builds confidence in Mathematics through structured guidance.
- **For Teachers:** Offers a data-driven dashboard to track class progress to proactively detect under-performing students, allowing for immediate interventions and academic decision-making based on predictive analytics rather than just traditional assessments.

1.8 Summary

In conclusion, in line with the advancement of the Fourth Industrial Revolution, this project aims to enhance the education capabilities of teaching the Mathematics subject to Form 4 and Form 5 KSSM students through the development of a Predictive Learning System. By analyzing students' behavioral and academic data, the system can predict the students' academic outcomes, provide personalized learning, and promote more efficient learning methods through the integration of machine learning techniques.

CHAPTER TWO

REVIEW OF PREVIOUS WORK

2.1 Introduction

The literature review helps to look at education today, where it focuses on finding the gap between teaching methods and what is needed in the Fourth Industrial Revolution (4IR). The main objective is to confirm that we should move from storing information to using AI to predict and improve learning environments. The Fourth Industrial Revolution needs a revolutionary advancement in the learning sector. Therefore, we analyze current learning systems to identify how to implement AI for improving current system frameworks.

This chapter discusses previous studies that act as the foundation to the development of predictive mathematics learning systems. The purpose of this chapter is to determine the functionality and gaps in current educational systems, and to see how far the integration of an intelligent AI model can improve the learning process, particularly for form 4 and 5 students in Mathematics.

2.2 Overview of Related Systems

Table 1 Comparison of Developed System

No	Article	Problem	Methods	Advantages	Limitation	Supporting Evidence
1	Villegas et. al. (2025)	Traditional systems lack real-time dynamic capability and traditional STEM teaching uses a standardized approach that fails to adapt to individual student needs.	Development of an AI-based Intelligent Tutoring System (ITS) integrating deep learning and Natural Language Processing (NLP).	Increased accuracy in solving mathematical problems and learning management	Relying on probabilistic models fails to achieve specific student needs	The experimental group achieved 85% precision in programming and 78% in mathematics (outperforming the control group). Additionally, 80% of students found the adaptive feedback highly useful.
2	Munassar dan AL-Hurdi (2025)	Previous models forecasting student performance frequently encounter difficulties with multi-class categorization, and tri-class data	Combines Random Forest and Deep Neural Network algorithms into a hybrid model.	Improves data mining accuracy and decision-making reliability.	Model performance deteriorates as the number of classes increases, especially when the boundaries between classes overlap.	The model successfully separated low, medium, and high performance with 80% accuracy. Furthermore, the hybrid model demonstrated its potential by achieving an accuracy of up to 96% in

		negatively impacts algorithm performance.				binary classifications.
3	Zulpykhar et al. (2025)	Traditional educational practices follow fixed structures that do not accommodate students with varying needs.	Two groups were studied, one using an adaptive platform and one following standard classroom instruction.	Personalised learning improved academic performance among underperforming students.	The small sample size limits the applicability of the results.	The group using the system scored higher on tests.
4	Lai, C.-H., & Lin, C.-Y. (2025).	Traditional programming education often delays students' problem-solving due to lack of timely help, which can adversely impact learning outcomes, especially for less confident students.	Uses an LLM-based module to generate responses that guide students without revealing direct answers.	Provides instant, unbiased assistance that enhances the continuity and efficiency of learning. It takes the load off the instructor and allows the student to receive individualized, step-by-step assistance rather than just being handed the answers.	The system does not track student progress or achievements.	Most students responded positively to the system.
5	Ramadhan, et al.	Classroom	The study conducted an	The system	The experiment relies	Quantitative data

	(2025).	instruction emphasises memorisation, which contributes to student disengagement.	experiment with 100 high school students using an adaptive system, split between a control group using traditional quizzes (n=50) and an experimental group using the digital platform (n=50).	dynamically adjusts question difficulty and delivers targeted, real-time feedback.	on a specific sample size of 100 high school students, which inherently limits the broader generalizability of the results across different age groups, subjects, or larger populations.	collected via pre- and post-tests showed a marked improvement in the experimental group ($p < 0.001$). This was further supported by qualitative insights gathered from student interviews and surveys.
6	Zhiyenbayeva et. al. (2021)	Existing educational frameworks rely on a "one-size-fits-all" approach that fails to adapt to individual student abilities, particularly disadvantaging learners who struggle with foundational baseline skills.	An experiment involving 78 engineering students utilizing a personalized, computer-supported learning system. The system assessed baseline mathematical abilities and tailored the educational process to match the students' unique learning trajectories.	Significantly enhances academic performance by actively adapting to individual student needs, proving that dynamic, customized support is more effective than traditional, static classroom instruction.	Identification of specific outcomes and aspects of this learning environment might be influenced by the waiting effect.	The impact of personalized learning on academic performance is moderately high, with a determinacy coefficient of 72.6%. Importantly, low and medium ability students saw more academic growth than high-aptitude students.
7	Alharbi &	Predicting academic performance is	Used a PHC combining algorithms in an optimal	Joint algorithms outperformed	The study relies on secondary data, which	The PHC outperforms other algorithms in

	Allohibi (2024)	difficult. Traditional single machine learning algorithms often find it hard to capture complex, nonlinear relationships in high-dimensional educational datasets.	configuration.	individual ones in predicting student performance.	may not reflect real-world conditions.	classification accuracy.
8	Yousafzai et. al. (2021)	Noisy data makes student performance difficult to predict.	Employed an improved feature selection technique followed by a deep neural network model	Automatically extracts key contextual elements from unprocessed education data to provide better deep learning abilities for sequencing and make very accurate early assessments of overall grade.	Many deep neural network models need a large amount of good quality training data to run at optimal levels; however, in many cases, the size of the dataset is small and there are few classification labels available (for example, binary pass/fail).	The performance of the proposed system achieves an accuracy of 90.16% which is better compared to existing machine learning techniques

2.3 Discussion

Studies of adjacent fields indicate that recent developments in Artificial Intelligence (AI), Intelligent Tutoring Systems (ITS), and Machine Learning (ML) have enhanced the learning experience. Most studies note that personalized learning is key to addressing challenges that most of the traditional teaching systems face, where all students are taught with the same learning pace and level irrespective of students' abilities and learning pace. Villegas et al. (2025), Zulpikhar et al. (2025) and Zhiyenbayeva et al. (2021) showed that the use of adaptive learning systems resulted in better student performance owing to personalized learning and feedback. Likewise, Lai and Lin (2025) noted that LLM tutoring systems provide on-the-spot support, and Ramadhan et al. (2025) noted that adaptive quizzes support learning and increase student participation. All these studies show that educational systems that use adaptive learning fosters a more engaging experience than the traditional classroom.

Furthermore, some limitations can be identified, even with the encouraging findings. Most of the educational support systems studied focus on one of the following components: personalized support, intelligent tutoring, or academic performance support, and do not combine all the components into one system. Munassar and Al-Hurdi (2025), Alharbi and Allohibi (2024), and Yousafzai et al. (2021) reached a high level of accuracy in predicting student performance, but did not incorporate adaptive learning and instructional support. There also were other limitations, including small sample sizes, limited access to data, and restricted evaluation methods that limit the findings' applicability. Therefore, a system that integrates all of the above is required. An integrated intelligent tutoring system with all of the above components could provide the educational support system that teachers need to better support their students.

The proposed system integrates Random Forest and Deep Neural Networks, which is influenced by the need for high accuracy in determining suitable questions according to students' academic mastery level. For the system development, we use the Iterative and Incremental Development Model because it is flexible, which allows for iterative enhancements for our AI chatbot and predictive algorithms based on user feedback, which is the students. In addition, the proposed system uses a performance dashboard to show student analytics to give insights to teachers if they want to implement an improvement in their

teaching methods. Compared to other related systems, which are mostly static instructions, the proposed system differentiates each question given with academic mastery level based on behavioral data like response time. While related systems usually use the same identical modules and questions, the proposed system adjusts the question complexity dynamically and integrates a scaffolding chatbot to assist students with hints without giving away the final answer. The proposed system also aligns with the KSSM Mathematics syllabus and creates a performance dashboard for teacher intervention. Therefore, the proposed system produces much more than related systems mentioned in the table by offering a more sustainable solution which can be applied by both students and teachers.

2.4 Summary

To conclude, the literature review proves that existing systems lack the real need for students' academic needs. Technologies like AI and IoT are proven to be able to be applied in the current learning systems to create a much more sustainable and adaptive environment to improve the overall education in Malaysia. With the current existing systems being only stationary and constantly using the same module, the proposed system is considered revolutionary for the education sector as it integrates the current technologies, such as AI and IoT. The results of our findings proposed the solution to the development of the proposed systems which make use of scattered information resources to be translated into a simplified smart learning system that uses machine learning algorithms.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter describes the methodology used in the development of the Predictive Learning System for Mathematics Form 4 and Form 5, utilizing Machine Learning (ML). This methodology is organized systematically to ensure that every development phase, starting from data preparation, model design, and system development, to performance evaluation, is executed effectively to achieve the research objectives. The first phase begins with requirements and data preparation, involving the collection and preparation of student performance data and a mathematics question bank. This process also includes Exploratory

Data Analysis (EDA) to understand data distributions and patterns. Research shows that when applied to educational data mining, predictive modeling coupled with EDA contributes greatly to personalized learning interventions (Baker & Inventado, 2014). EDA is typically used as the primary data analysis method to comprehend the distributions and patterns of the data. McKinney (2022), states that in a machine learning pipeline, the absence of effective exploratory data analysis can lead to problems caused by the data's hidden distributions and anomalies, which can significantly affect the model's performance after it is trained.

Once the data is prepared, this chapter justifies the selection of the Hybrid RF-DNN model used to classify student mastery levels more accurately. Studies investigating different algorithms highlight that incorporating ensemble methods, especially Random Forest, and Deep Neural Networks, optimizes classification using advanced feature selection (Adefemi et.al., 2025). Furthermore, this chapter discusses the integration of a chatbot utilizing a scaffolding approach. This chatbot functions to support students through step-by-step hints without providing direct answers, helping students think actively while solving mathematical problems. Winkler, R., & Söllner, M. (2018) suggests that the addition of large language models as teaching agents can help become a scaffold, and for example, help students become more cognitively engaged as opposed to answer-seeking. Finally, this chapter explains the experimental specifications, including data splitting, hyperparameter tuning, and model performance evaluation metrics. System specification requirements are also discussed, covering functional, hardware, and software aspects to ensure the system operates smoothly in the subsequent phases.

3.2 Development Approach

This project is implemented using the Iterative and Incremental Development Model. In educational software engineering, it creates opportunities for continual stakeholder alignment and diminishes the potential for technical risks throughout the development lifecycle (Larman & Basili, 2003). This approach was selected due to its flexible nature, where the system is developed through a series of repeating cycles that allow for continuous improvements until the desired level of quality is achieved. This methodology is highly effective for the development of the Hybrid RF-DNN model and chatbot integration, as it provides the necessary space to perform fine-tuning at each phase based on testing results and feedback received.

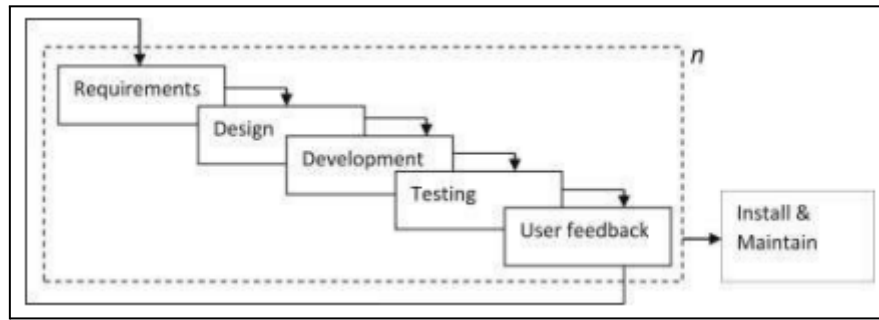


Fig Iterative and Incremental Development Model Diagram

The first phase, Requirements, involves gathering critical technical and academic information to determine the system's development direction. Key activities include the analysis of secondary datasets from the Kaggle repository to identify student performance profiles, as well as the organization of a Mathematics question bank aligned with the Form 4 and Form 5 KSSM syllabus. The output of this phase is a clear list of functional requirements, including the criteria for classifying student mastery into three levels: Low, Medium, and High.

The second phase, Design, involves the detailed development of the system architecture and the machine learning model framework. This includes constructing the application workflow structure using the Streamlit platform, mapping question difficulty levels against model outputs, and designing the system prompts for the guidance chatbot. Python-based frameworks such as Streamlit provide a way for developers to rapidly create interactive and user-friendly web applications from their machine learning prototype (Treveil et al., 2020). This design planning is vital to ensure that the chatbot provides appropriate and systematic support to help students understand mathematical concepts without directly disclosing the answers.

The third phase, Development, focuses on writing programming code using the Python language to build system modules incrementally. Key processes include data preprocessing, training the hybrid model which combines RF-DNN algorithms, and integrating LLM APIs to activate virtual tutor functionality. Kotsiantis et al. (2006) states that, a basic prerequisite for high fidelity in neural network architectures is proper data preprocessing, including normalization and missing value imputation. Development is conducted modularly so that the adaptive quiz component functions stably before being fully integrated with other artificial

intelligence components.

The fourth phase, Testing, is carried out comprehensively for each completed module to evaluate the system's stability and accuracy. Primary focus is placed on assessing the accuracy metrics of the classification model and the effectiveness of the adaptive logic in generating questions according to the detected student competency levels. The importance of rigorous evaluation of the model, by means of such criteria as accuracy, recall, and F1-score, cannot be overlooked (Hossin & Sulaiman, 2015). This testing aims to detect any programming errors or logical failures in the delivery of mathematical content before the system moves to the user review stage.

The fifth phase, User Feedback, is a core element of this model where the system is reviewed by supervisors or subject matter experts to obtain qualitative evaluations. Feedback regarding the accuracy of student classification and the effectiveness of chatbot guidance is collected to identify any performance gaps that must be addressed. If improvements are necessary, the workflow cycles back to earlier phases to update the model or question content, ensuring the final product quality meets project objectives.

The sixth phase, Install & Maintain, involves launching the complete adaptive learning system as a proof-of-concept for educational use. Comprehensive technical documentation is prepared, covering model performance reports and system user manuals to ensure it is ready for use or further expansion by other researchers in the future. Periodic maintenance is also performed on the integrity of the question database and chatbot API connectivity to guarantee the holistic sustainability of system functions.

3.3 Requirements Specification

To develop a comprehensive smart learning ecosystem, this study integrates two types of datasets acquired through different procurement strategies, each playing a specific role within the system structure.

The first dataset, used as the primary training data, is the Student Performance and Learning

Style Dataset obtained from the open data repository, Kaggle. This dataset contains 14,003 student profile records covering various variables such as demographics, academics, and psychology. The selection of this large-scale dataset aligns with the requirements of Deep Learning algorithms, which necessitate substantial data volumes to prevent overfitting and enhance predictive accuracy. According to Sarker (2021), neural network models require extensive data to learn complex patterns effectively, thereby increasing predictive precision when tested against new data.

The second dataset is the Mathematics Question Bank. The data acquisition process for this dataset was conducted entirely manually without the use of any automation tools or web scraping. An initial 500 questions were entered into the database. Slater (2025), suggest that a sample size of 500 is sufficient for most cases of model construction, for a wide range of different algorithm types, with models intended to generalize to new populations of learners benefitting from slightly smaller sample sizes. The questions are divided into low, medium or high difficulty with validation from subject matter experts mainly teachers who teach Mathematics. The time allocated for each question is based on the difficulty given, 60 seconds for low, 90 seconds for medium and 120 seconds for high. During the prototype phase, the question bank began with a small volume to test system flow; however, to ensure comprehensive content, this bank must be expanded further by topic and difficulty level in future phases.

The screening, selection, and data entry processes were meticulously carried out by referring to authentic educational sources, such as the Bumi Gemilang (2024) books, which comply with the Mathematics KSSM Form 4 and Form 5 Curriculum and Assessment Standard Document (DSKP) specifications issued by the Ministry of Education Malaysia (2018). Manual curation and expert validation of educational datasets are still important to ensure content validity and adherence to standardized curriculum frameworks (Chounta & Carvalho, 2022). Each question was typed individually into the database along with essential metadata such as topic, difficulty level, hints, and allocated answering time. The difficulty levels were determined with guidance from experts, specifically school mathematics teachers who instruct Form 4 and 5 students. This manual data acquisition approach was adopted to ensure content quality and question validity, while guaranteeing a data format consistent with the adaptation logic requirements of the developed system.

System requirements define the necessary properties of the system with each function playing an important role to ensure the system delivers a personalized and elective learning experience for the students.

3.3.1 Functional Requirements

- Allow students to log into the system to access their personal learning profile
- Provide an input field for students to enter academic profile data such as past exam scores, study time, attendance, and internet access level.
- Process student input data to classify the student's initial mastery level (Low, Medium, or High).
- Automatically generate a set of customized Mathematics quiz questions based on the classified mastery level.
- Display a quiz interface with a timer running to record the duration of answering each question.
- Dynamically adapt the difficulty level of the question (lower the level/remain at the same level) if the student takes more time to answer than the set limit.
- Provide interactive assistance through a Guidance Chatbot that provides hints or guidance without providing a final answer.
- Calculate the student's quiz score and apply penalties (mark deductions) automatically each time the student uses the Chatbot's assistance.
- Store historical records of student quiz results into a database to enable monitoring of trends over time.
- Display graphical visualizations (such as charts or graphs) that show the development of student mastery levels.

3.3.2 Non-Functional Requirements

- Seamlessly synchronize the user interface with the backend, process data, run quizzes, and facilitate real-time interactions with the chatbot.
- Maintain stable connectivity between the Streamlit frontend, the ML backend, and the LLM chatbot API.
- The Hybrid RF-DNN model must achieve a sufficiently high classification accuracy in predicting student mastery levels.

- Must process student responses and classify mastery levels in real time

3.4 Logical Design

Figure details the system operation sequence that begins with the user login process and the collection of initial profile data, which is then fed to the data pre-processing module for standardization. The next flow involves the decision phase where the RF-DNN Hybrid model analyzes the data to classify the student's mastery level into certain categories. Based on the results of this classification, the system will automatically perform a content selection process to display a set of questions that match the student's difficulty level, either a basic set or a reinforcement set. This process continues with student interaction to answer the questions, where they can access Chatbot assistance if necessary, and ends with the score evaluation and updating of the performance record into the database before the system decides whether to proceed to the next topic or end the session.

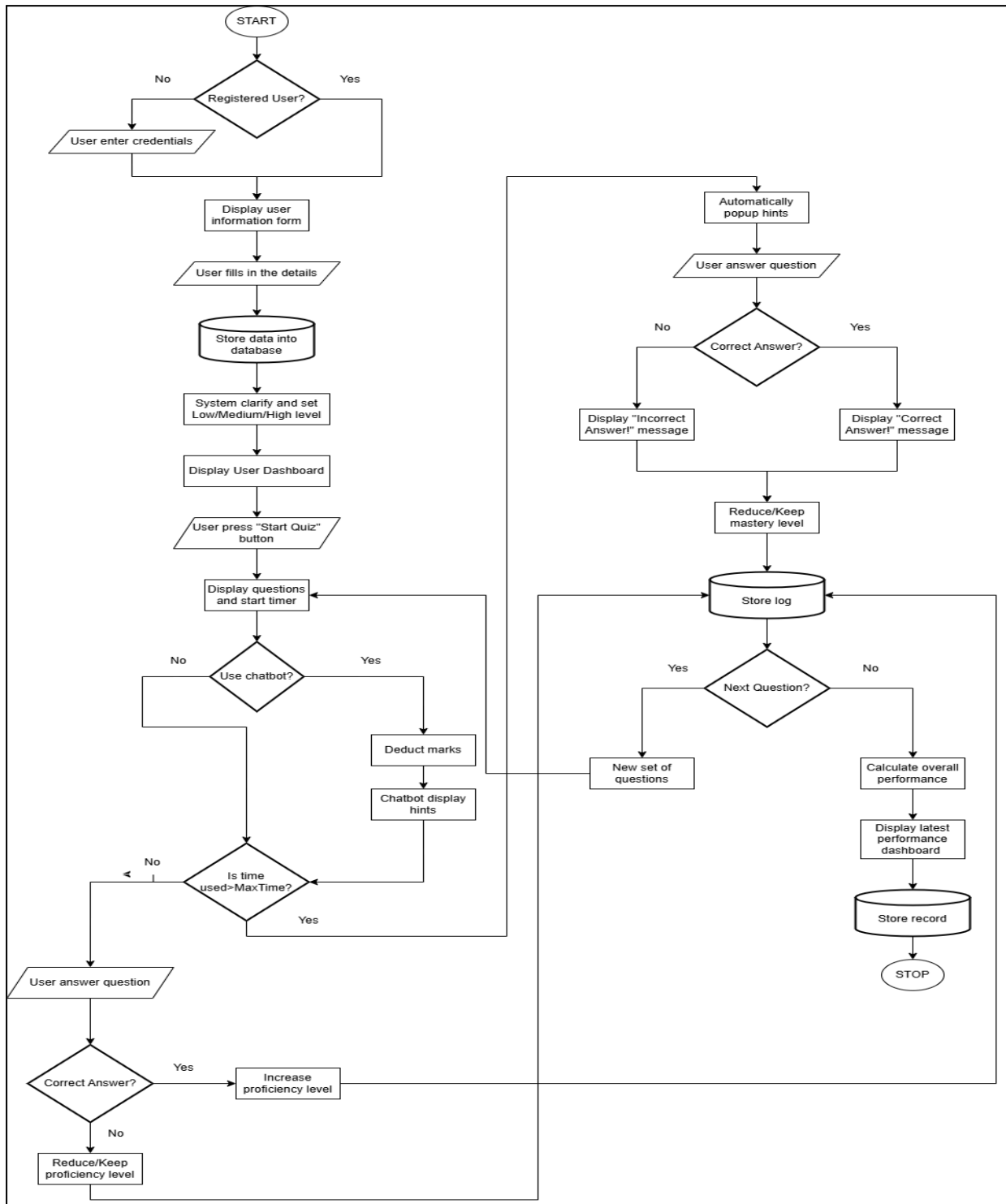


Fig Flow Chart Diagram

Figure visualizes the interaction between the User actor and the main functions in the system, starting with the authentication phase where the relationship (<<include>>) between the use cases 'Enter profile credentials' and 'Login account' establishes that logging in is an essential step before the profile can be accessed. The core function 'Answer Adaptive Quiz' starts with 'Mastery Level Classification' (<<include>>) to determine the student's initial

competence, while the adaptation mechanism is handled through the relationship (<<extend>>) on the 'Update Question Level' function. In this process, if the student is detected to be taking too long, the system will automatically display a hint, and immediately after the answer is submitted, the system will provide performance feedback and activate the 'Update Question Level' function to maintain or lower the difficulty level for the next question. Conversely, if the student successfully answers correctly within the allotted time, the system will increase the question level to a higher level. In addition, students can choose the 'Ask for chatbot help' option (<<extend>>), but it is subject to the condition of 'Count marks deduction' (<<include>>) being implemented immediately, and finally, all achievement patterns can be monitored through the 'View performance analytics' function.

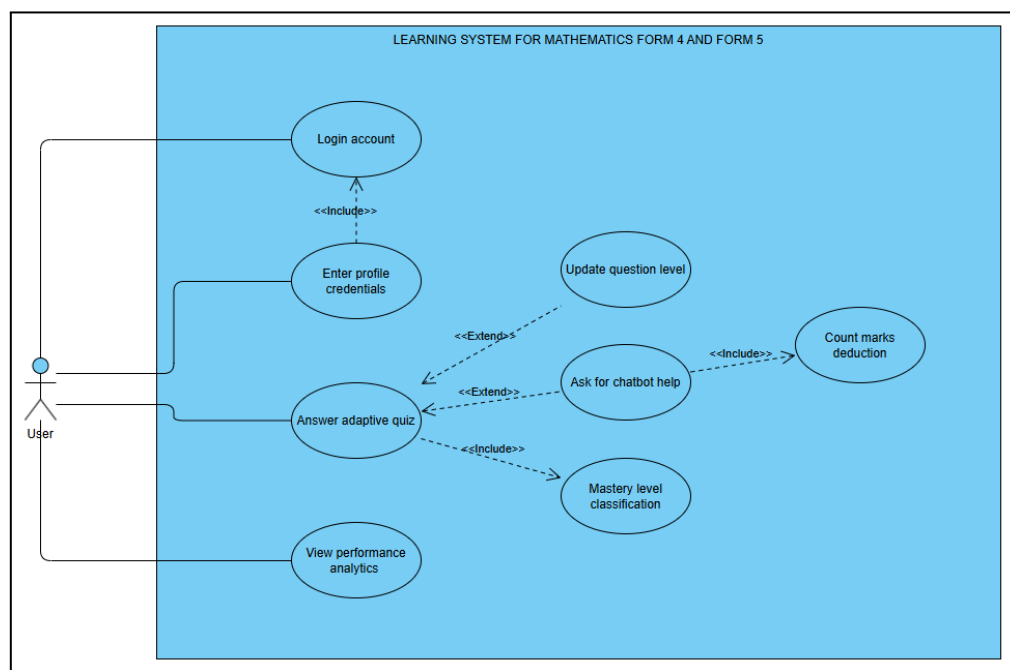


Fig Use Case Diagram

Figure shows a structure divided into four main layers that interact with each other, starting with the Data Layer, which functions as a repository for storing the KSSM Mathematics question bank, student profile records, and historical performance data. Raw data from this layer is then fed to the Intelligent Processing Layer, where the Hybrid Random Forest and Deep Neural Network (RF-DNN) artificial intelligence model is trained using balanced data through the SMOTE technique, which analyzes the input to predictively classify the student's mastery level. The classification results are then applied by the Application Logic Layer, which houses adaptive algorithms to dynamically determine the difficulty level of the next

question and manages API integration for the Guidance Chatbot function. Finally, all these backend processes are translated into a visual experience through the Presentation Layer, which is developed using the Streamlit framework, providing an interactive user interface for students to answer quizzes and receive real-time performance feedback.

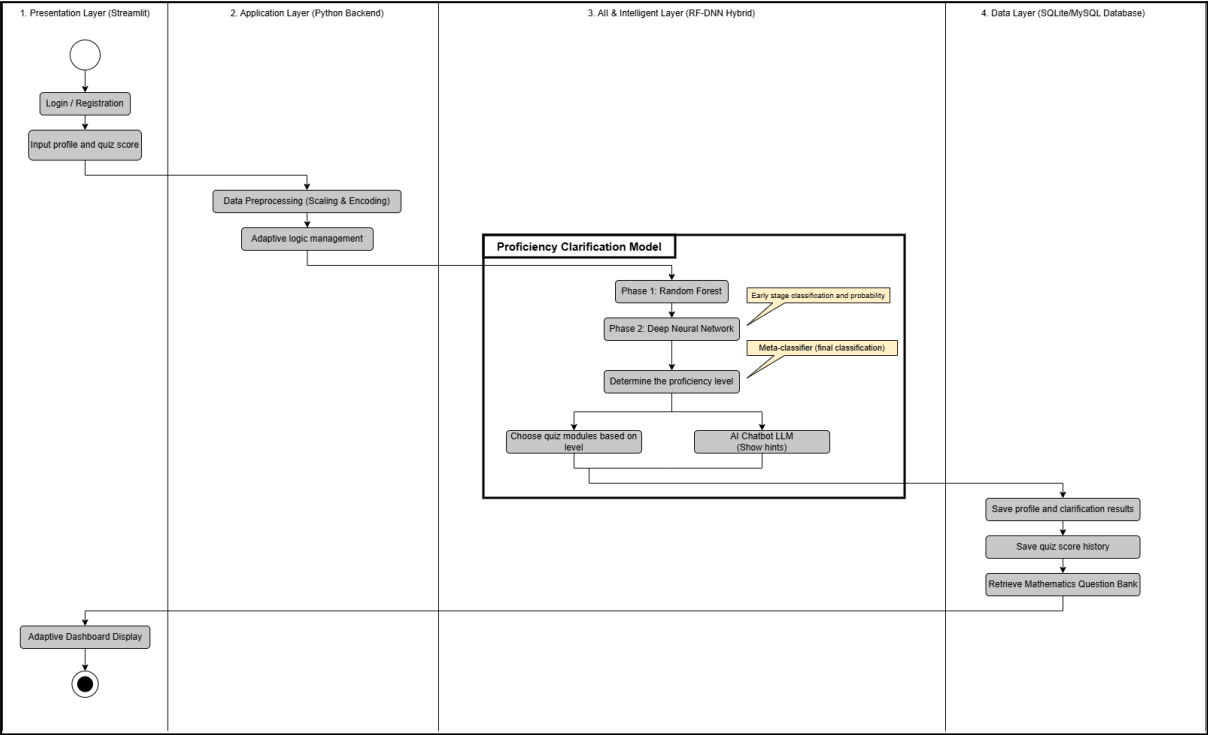


Fig Activity Diagram

Figure shows the class diagram, which maps the structural relationships between the core entities in the Predictive Learning System. The 'Student' class is the primary actor, answering quiz modules and viewing performance analytics. The 'Quiz' class is the central component, containing attributes such as question difficulty, topic tags, and metadata used for dynamic adaptation. The 'Chatbot' class provides real-time scaffolding support through hint generation, while the 'Performance' class tracks metrics and updates the student's profile. The Hybrid Model class processes the student's initial input data alongside performance analytics to predict mastery level.

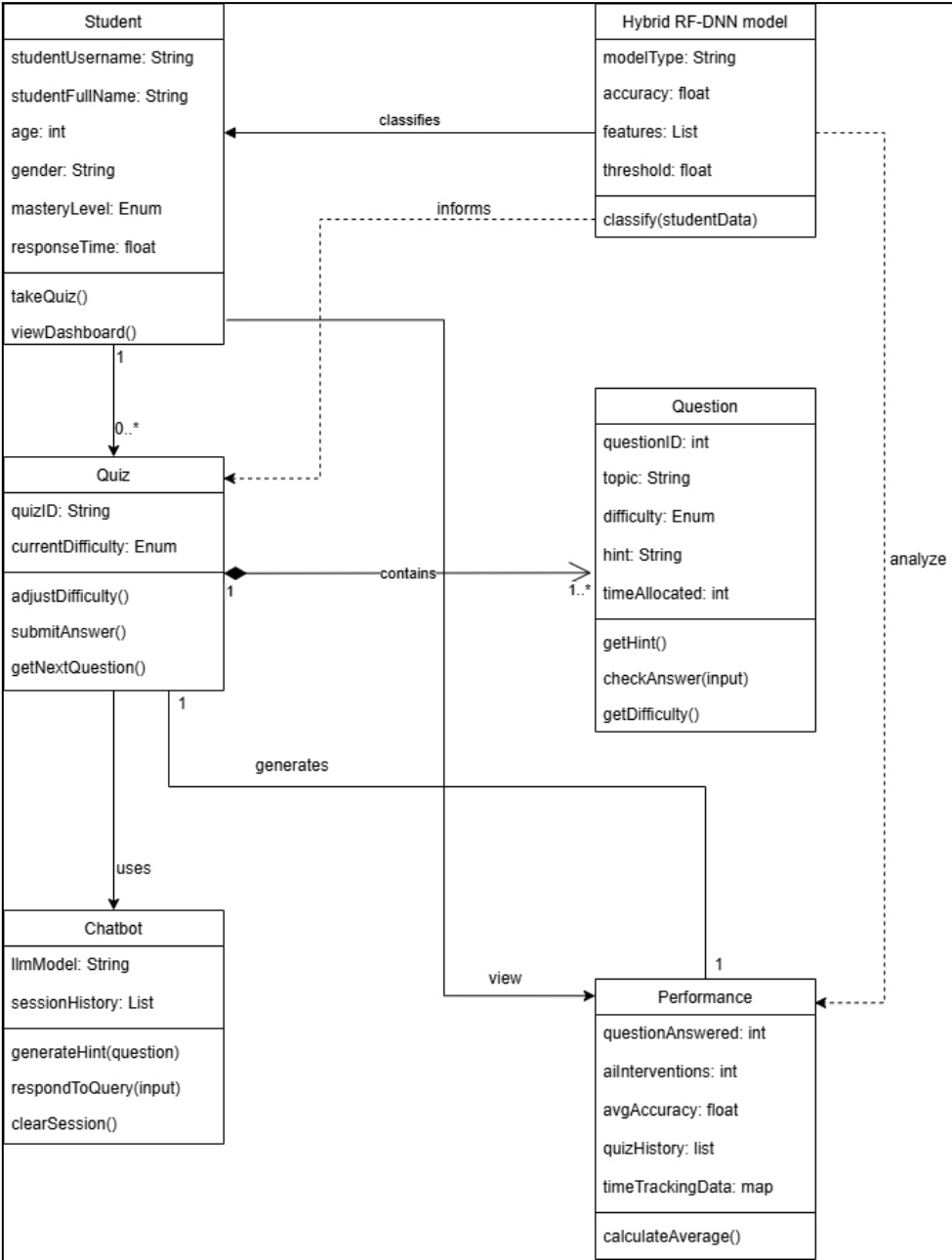


Fig Class Diagram

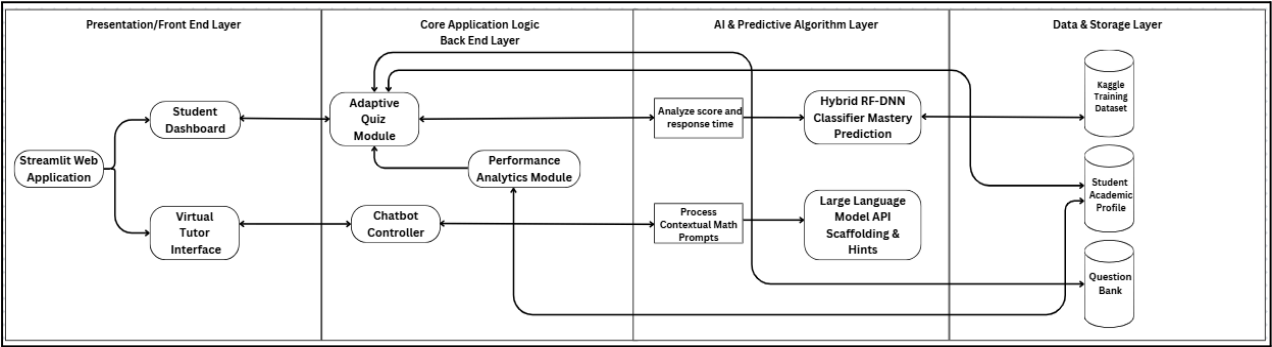


Fig Architecture Diagram

3.5 Database Design

Figure shows the Entity-Relationship Diagram (ERD), which maps the logical database structure and relational data flow between the core entities in the Predictive Learning System. The 'STUDENT' entity is the central repository for user profiles, storing demographic details alongside the predictively generated mastery level. A student interacts with the assessment module through a 'QUIZ_SESSION', which records the link between the student and a specific assessment instance, including the date taken and current difficulty level. Each session contains items from the 'QUESTION' entity, which stores a questionID, topic, difficulty level, hint message, and allocated time per question. Upon completion, each quiz session produces exactly one 'PERFORMANCE' record, storing metrics such as average accuracy, frequency of AI interventions, and time-tracking data that the Hybrid RF-DNN model uses for mastery prediction. Finally, the CHATBOT_LOG entity tracks scaffolding interactions initiated by the student, referencing the specific question being attempted and recording the student's input query, the system's generated response, and a timestamp. This allows learning behaviors to be reviewed and confirms that the AI provides guidance without revealing direct answers.

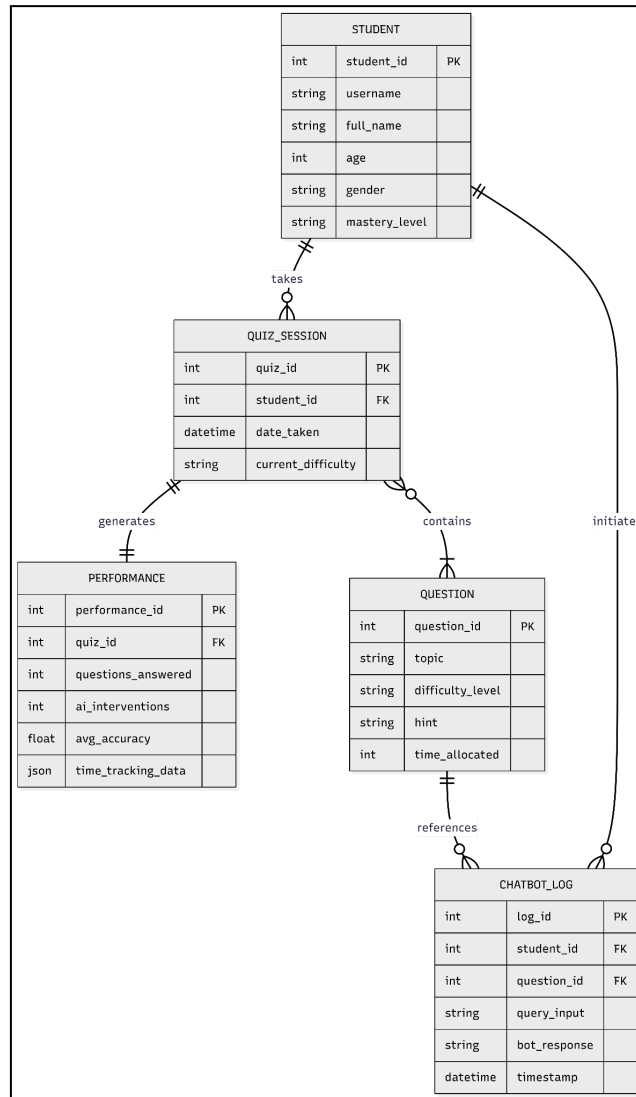


Fig Entity-Relationship Diagram

3.6 Prototype or Wireframes

KICT | Adaptive Math System

Welcome Back

Log in to continue your adaptive learning journey.

Username

Password

LOGIN >>

Don't have an account? [Register here](#)

Fig Login Page

KICT | Adaptive Math System

Account Details

Choose a Username

Choose a Password

Full Name

Select Gender

Select Age

Academic Profiling

Latest Exam Score (%)

State of Homework

Attendance (%)

Current Motivation State

REGISTER & START >>

Already have an account? [Log in here](#)

Fig Registration Page

Developing high-fidelity prototypes early in the design phase significantly improves user interface usability and overall learner satisfaction in digital education platforms (Walker et al., 2002). This section of the prototype covers the login and sign-up pages. Unlike a standard registration form, the sign-up page collects academic and behavioral data from the student, including exam scores, homework completion rates, class attendance, and motivation levels. Once this data is submitted, the prototype automatically assigns a starting mastery level of Low, Moderate, or High before the student enters the main application.

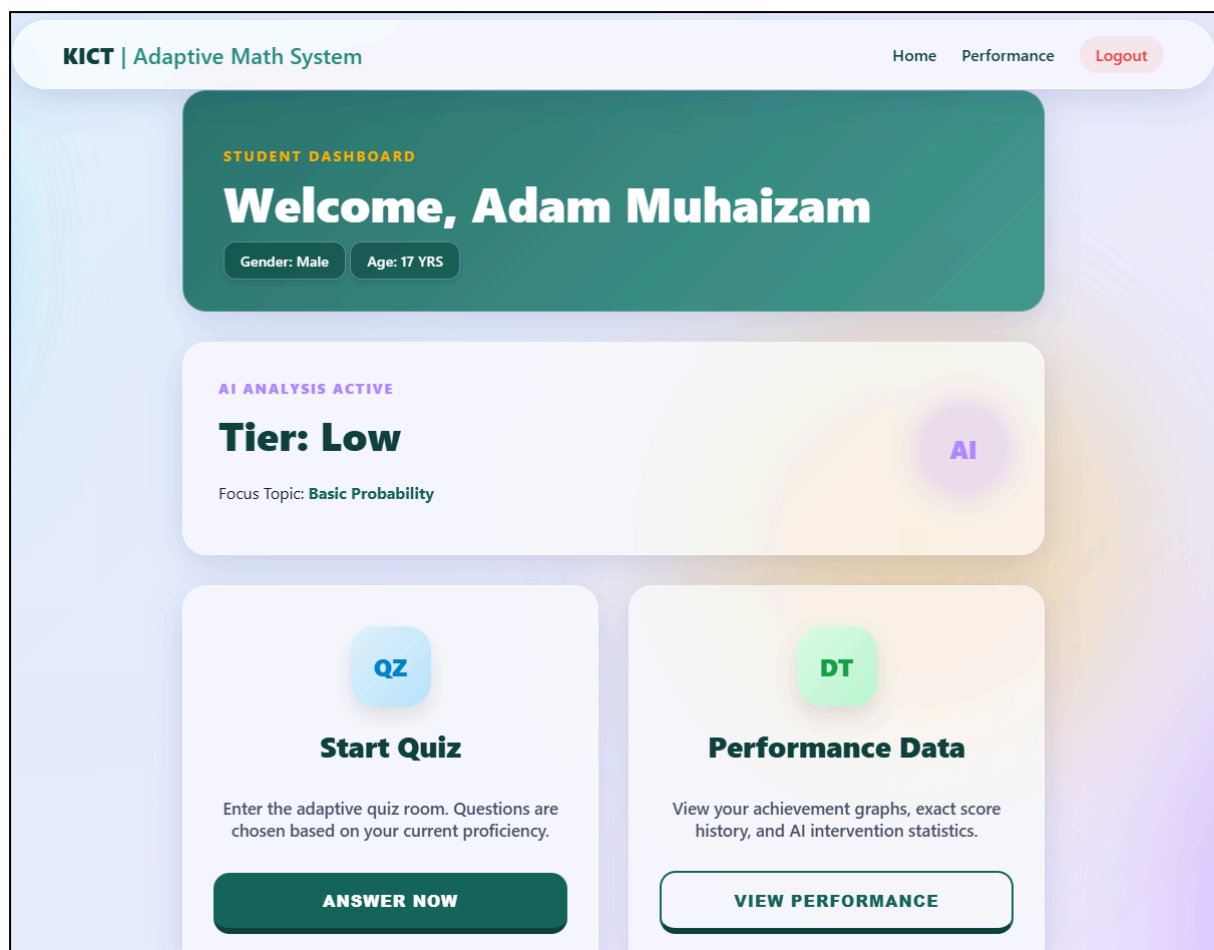


Fig Student Dashboard Page

The Student Dashboard page is the main navigation page. A gamified welcome banner sits at the top, displaying demographic information and profile tags. Below it, an "AI Analysis" card shows the student's current proficiency tier and focus topic, which the system updates dynamically. From this page, students access two action buttons: one to launch an adaptive quiz and one to view their performance history.

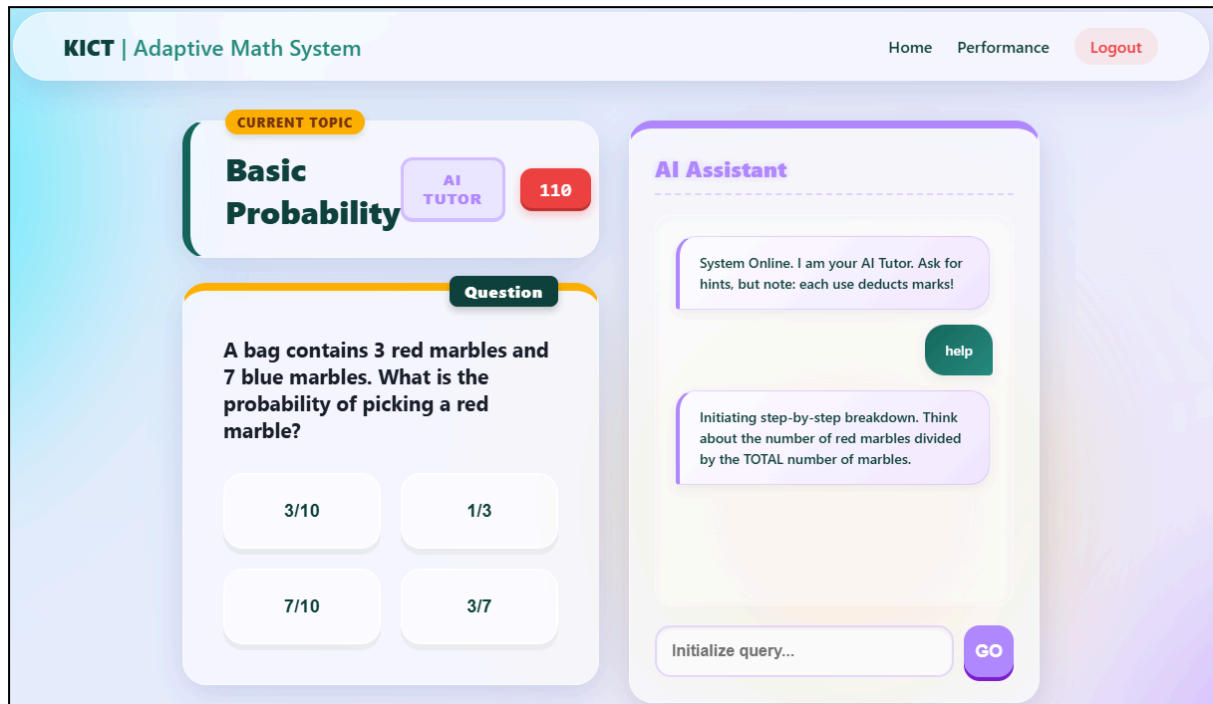
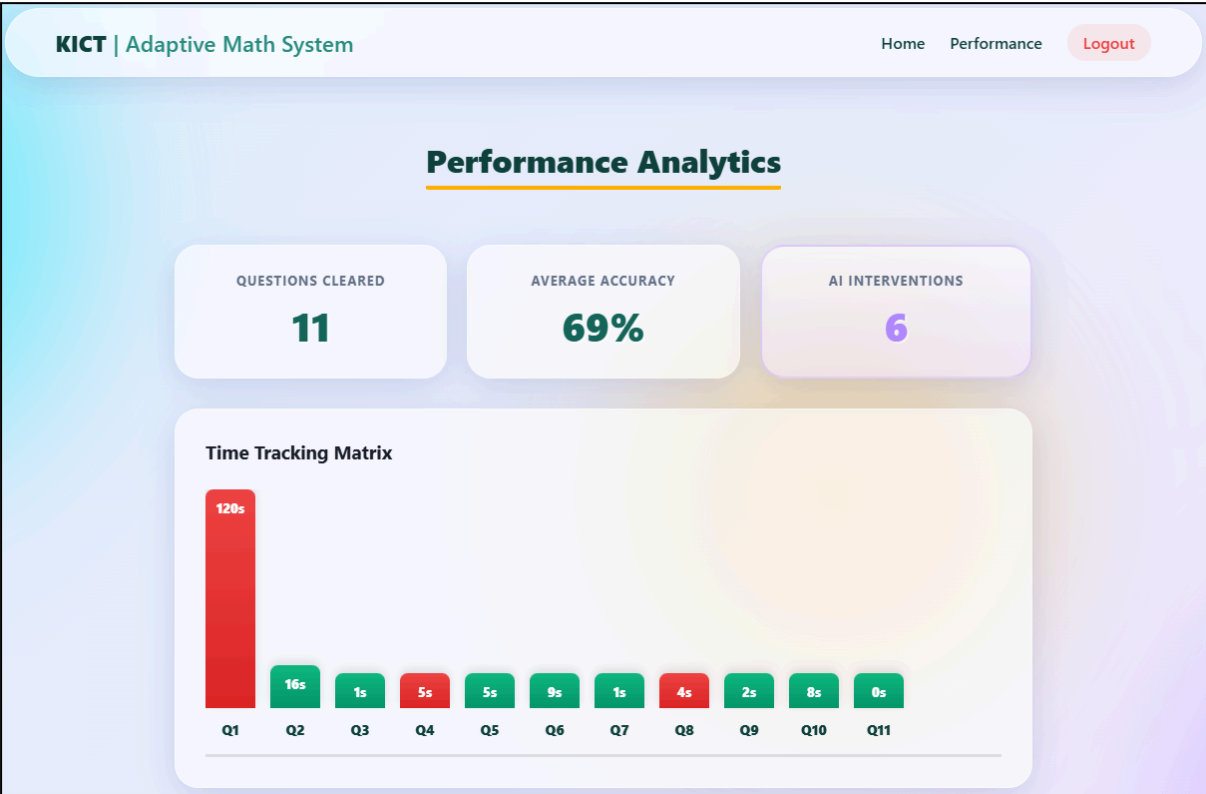
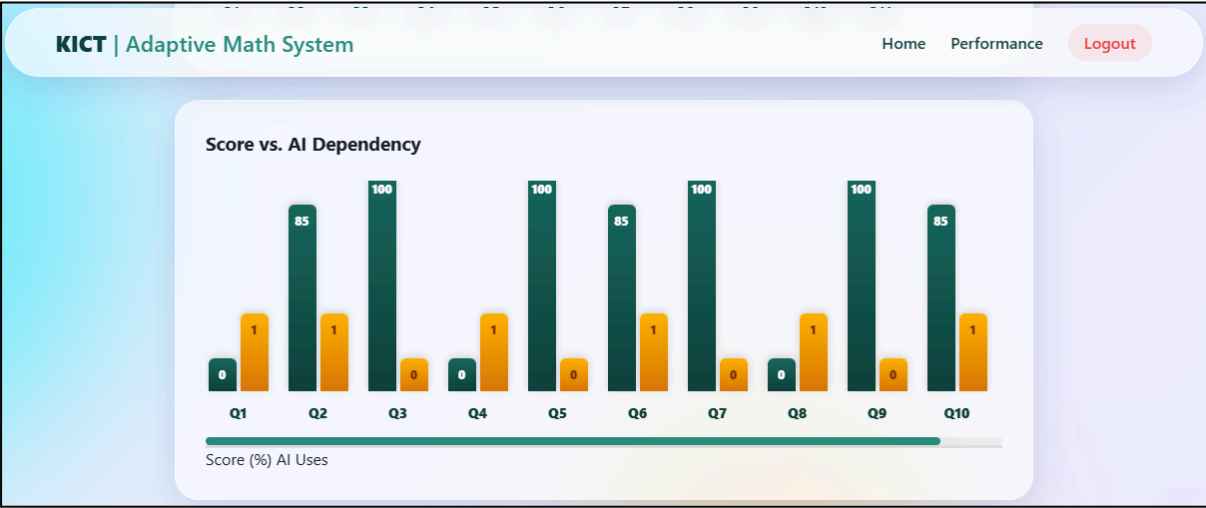


Fig Adaptive Quiz and AI Tutor Page

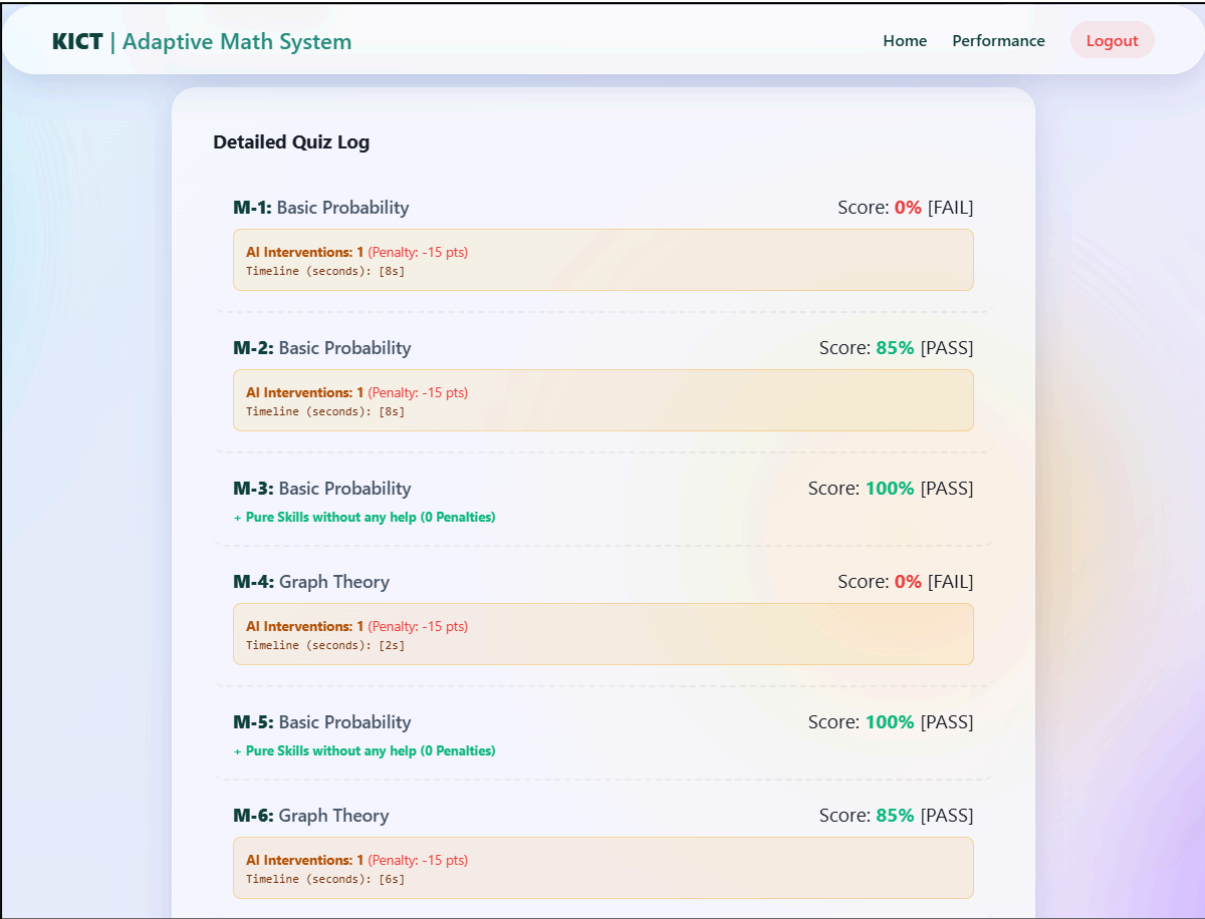
The Adaptive Quiz page is the core testing environment, integrating a question interface with an AI Tutor. A question box displays mathematics questions matched to the student's current level, while the Deep Neural Network (DNN) algorithm selects the next question based on performance. To progress to a higher difficulty level, the student must maintain an average score at the current level. A countdown timer below each question records response speed. Next to the question box, a slide-out AI Tutor chatbot allows students to request hints. Each request incurs a point deduction, discouraging overreliance on AI assistance and promoting independent critical thinking.



Time Tracking Analytics Page



Average Score and AI-dependency Page



Detailed Quiz Log Page

The Performance Analytics page displays a detailed breakdown of the student's previous data and learning path. The first section summarizes total questions answered, average accuracy, and number of AI interventions, presented as text. Below this, two charts present the data visually: a Time Tracking Matrix showing average response time, and a Dual Bar chart showing the student's score alongside their Chatbot AI Intervention Score. The final section lists all completed quiz sessions in chronological order, including the score, response time, and penalties from chatbot hint usage.

3.7 Summary

To conclude, the methodology that has been used for the Predictive Learning System is to create a robust framework in creating an adaptive learning environment while also providing flexibility with iterative cycles to allow the tuning of the system including scaffolding AI chatbox. The six phases mentioned in the Iterative and Incremental Model consist of specific requirements gathering until installation and maintenance are implemented

to meet feedback expectations. The strategy also provides a reliable system to implement a large-scale dataset that consists of 14,003 student records from Kaggle and the mathematics question bank. These resources align with the KSSM syllabus which means this system can be applied to Form 4 and Form 5 students in Malaysia. Overall, this approach transforms complicated student data into a personalized and interactive learning experience.

REFERENCES

- Kementerian Pendidikan Malaysia. (2025). Laporan analisis keputusan SPM 2024. Lembaga Peperiksaan Malaysia.
<https://lp.moe.gov.my/files/spm/2025/Laporan%20Analisis%20Keputusan%20SPM%202024.pdf>
- Galaige, J., & Torrissi-Steele, G. (2017). 'One size does not fit all': Towards cultural adaptivity in learning management systems. *ASCILITE 2017 Conference Proceedings: Me, Us, IT!*, 78–82. <https://doi.org/10.14742/apubs.2017.746>
- Valdemoro, L. A. P., Besa, E. B., Lim, M. D. R., Villanueva, A. L. O., & De Luna, J. N. N. (2026). Personalized learning path recommendation for system integration and architecture through a decision tree-based AI engine. *International Journal for Multidisciplinary Research*, 8(2). <https://doi.org/10.36948/ijfmr.2026.v08i02.72258>
- Villegas, William & Buenano-Fernandez, Diego & Navarro, Alexandra & Mera-Navarrete, Aracely. (2025). Adaptive intelligent tutoring systems for STEM education: analysis of the learning impact and effectiveness of personalized feedback. *Smart Learning Environments*. 12. 10.1186/s40561-025-00389-y.
- Mannekote, A., Davies, A., Pinto, J. D., Zhang, S., Olds, D., Schroeder, N. L., Lehman, B., Zapata-Rivera, D., & Zhai, C. (2024). Large language models for whole-learner support: Opportunities and challenges. *Frontiers in Artificial Intelligence*, 7. <https://doi.org/10.3389/frai.2024.1460364>
- Davar, N. F., Dewan, M. A. A., & Zhang, X. (2025). AI Chatbots in Education: Challenges and Opportunities. *Information*, 16(3), 235. <https://doi.org/10.3390/info16030235>
- Lin, C.-C., Huang, A. Y. Q., & Lu, O. H. T. (2023). Artificial intelligence in intelligent tutoring systems toward sustainable education: A systematic review. *Smart Learning Environments*, 10(1), Article 41. <https://doi.org/10.1186/s40561-023-00260-y>
- Leddo, J., & Garg, K. (2021). Comparing the effectiveness of ai-powered educational software to human teachers . *International Journal of Social Science and Economic Research*, 6(3), 953–963. <https://doi.org/10.46609/ijsser.2021.v06i03.015>
- Son, T. (2024). Intelligent tutoring systems in mathematics education: a systematic literature review using the substitution, augmentation, modification, redefinition model. *Computers*, 13(10), 270. <https://doi.org/10.3390/computers13100270>
- Munassar, N. M. A., & Al-Hurdi, A. S. (2025). Enhancing students' academic performance classification in e-learning using hybrid model (Random Forest and Deep Neural Network). *Journal of Science and Technology*, 30(7), 86–98.

<https://doi.org/10.20428/jst.v30i7.2948>

- Zulpykhar, Z., Kariyeva, K., Sadvakassova, A., Zhilmagambetova, R., & Nariman, S. (2025). Assessing the effectiveness of personalized adaptive learning in teaching mathematics at the college level. *International Journal of Engineering Pedagogy (iJEP)*, 15(4), 4–22. <https://doi.org/10.3991/ijep.v15i4.52797>
- Lai, C.-H., & Lin, C.-Y. (2025). Analysis of Learning Behaviors and Outcomes for Students with Different Knowledge Levels: A Case Study of Intelligent Tutoring System for Coding and Learning (ITS-CAL). *Applied Sciences*, 15(4), 1922. <https://doi.org/10.3390/app15041922>
- Ramadhan, S., Atmazaki, A., Ningsih, A. G., Hayati, Y., Henanggil, M. D. F., Nursaid, N., Rahman, F., & Ghaluh, B. M. (2025). Exploring the impact of adaptive real-time quiz platforms with differentiated learning features on student engagement and learning outcomes: A mixed-methods approach. *International Journal of Information and Education Technology*, 15(6), 1261–1276. <https://doi.org/10.18178/ijiet.2025.15.6.2329>
- Zhiyenbayeva, N., Belyanova, E., Petunina, I., Dmitrichenkova, S., & Dolzhich, E. (2021). Personalized Computer Support of Performance Rates and Education Process in High School: Case Study of Engineering Students. *International Journal of Engineering Pedagogy (iJEP)*, 11(2), pp. 135–153. <https://doi.org/10.3991/ijep.v11i2.19451>
- Alharbi, A. A., & Allohibi, J. (2024). A new hybrid classification algorithm for predicting student performance. *AIMS Mathematics*, 9(7), 18308–18323. <https://doi.org/10.3934/math.2024893>
- Yousafzai, B. K., Khan, S. A., Rahman, T., Khan, I., Ullah, I., Ur Rehman, A., Baz, M., Hamam, H., & Cheikhrouhou, O. (2021). Student-Performulator: Student academic performance using hybrid deep neural network. *Sustainability*, 13(17), Article 9775. <https://doi.org/10.3390/su13179775>
- Baker, R. S., & Inventado, P. S. (2014). Educational data mining and learning analytics. In J. A. Larusson & B. White (Eds.), *Learning analytics* (pp. 61–75). Springer. https://doi.org/10.1007/978-1-4614-3305-7_4
- McKinney, W. (2022). *Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter* (3rd ed.). O'Reilly Media, Inc.
- Adefemi, K. O., Mutanga, M. B., & Jugoo, V. (2025). Hybrid Deep Learning Models for Predicting Student Academic Performance. *Mathematical and Computational Applications*, 30(3), 59. <https://doi.org/10.3390/mca30030059>
- Winkler, R., & Söllner, M. (2018). Unleashing the potential of chatbots in education: A

- state-of-the-art analysis. *Academy of Management Proceedings*, 2018(1), Article 15903. <https://doi.org/10.5465/AMBPP.2018.15903abstract>
- Larman, C., & Basili, V. R. (2003). Iterative and incremental developments: A brief history. *Computer*, 36(6), 47–56. <https://doi.org/10.1109/MC.2003.1204375>
- Treveil, M., Ojala, N., Marttinen, M., Kettunen, V., Vaananen, T., Hamalainen, H., ... & Laine, T. (2020). *Introducing MLOps: How to scale machine learning in the enterprise*. O'Reilly Media, Inc.
- Kotsiantis, S. B., Kanellopoulos, D., & Pintelas, P. E. (2006). Data preprocessing for supervised learning. *International Journal of Computer Science*, 1(2), 111–117.
- Hossin, M., & Sulaiman, M. N. (2015). A review on evaluation metrics for data classification evaluations. *International Journal of Data Mining & Knowledge Management Process*, 5(2), 1–11.
- Sarker, I. H. (2021). *Deep Learning: A Comprehensive Overview on Techniques, Taxonomy, Applications and Research Directions*. MDPI AG. <https://doi.org/10.20944/preprints202108.0060.v1>
- Slater, S. A. (2025). *Data quantity and data characteristics for modeling approaches in educational data mining* [Doctoral dissertation, University of Pennsylvania]. ScholarlyCommons. <https://repository.upenn.edu/entities/publication/355ca806-7b56-402b-97ae-17e4314c0c76>
- Chounta, I. A., & Carvalho, L. (2022). The role of expert curation in creating high-quality datasets for educational machine learning. *Computers and Education: Artificial Intelligence*, 3, Article 100065. <https://doi.org/10.1016/j.caeai.2022.100065>
- Walker, M., Takayama, L., & Landay, J. A. (2002). High-fidelity or low-fidelity, paper or computer? Choosing attributes when testing web prototypes. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 46(7), 661–665. <https://doi.org/10.1177/154193120204600707>
- Shamim, A. (2023). *Student performance and learning style* [Data set]. Kaggle. <https://www.kaggle.com/datasets/adilshamim8/studentperformance-and-learning-style>
- Bumi Gemilang. (2024). *Koleksi soaln percubaan SPM Matematik Tingkatan 4 & 5*. <https://www.bumigemilang.com/>

APPENDICES

A. GANTT CHART

PHASE	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Date	10-Oct	17-Oct	24-Oct	31-Oct	7-Nov	14-Nov	21-Nov	28-Nov	5-Dec	12-Dec	19-Dec	26-Dec	9-11 Jan	18-Jan
Data Pre processing														
Data Balancing														
Summarization of current FYP														
Data modelling														
Data training														
Model testing														
Website development														
Technical and final report														
FYP showcase														
Final report submission														

B. DIVISION OF WORK

A breakdown of each group member's tasks and responsibilities.

Name	Task
Mohammad Harith bin Azhan	- Project Planning and Requirement Analysis - Data Preprocessing and Balancing - Machine Learning Model Development - Performance Testing - Literature Review
Muhammad Afiq bin Ahmad Nizam	- Web Application Development - Question Bank Curation - Intelligent Chatbot Integration - Secondary Data Sourcing - Methodology